# Isais Marrufo

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## EDUCATION

**University of Southern California |** BS Computer Engineering & Computer Science **Graduation Date: Dec 2027**
GPA: 3.5/4.0

## EXPERIENCE

**Butte College** *August 2022 – August 2025*
*School Certifying Official – Assistant* *Oroville, CA*

- Managed and certified veteran student enrollments for 150+ students, ensuring compliance with VA guidelines
- Maintained and updated veteran student records using Enrollment Manager software
- Guided veteran students through the class registration process, course requirements, and schedules
- Developed and updated procedural manuals for veterans using Colleague software

**United States Army - Various Locations** *January 2018 – June 2022*
*Supervisor, Vehicle Recovery/Repair* *Fort Riley, KS*

- Led a team of 10 mechanics and oversaw service and recovery operations on 51 heavy tracked and light vehicles
- Provided technical guidance and mentored team members on proper troubleshooting procedures in compliance with Technical Manual 9-2350-292, researched Class IX repairable parts, and coordinated timely parts requisition
- Filled mission critical vacant supervisor role, increased, and maintained operational readiness of 4 Recovery Vehicles
- Accounted for 5 General Mechanics Tool Kits and 2 M88A2 Heavy Recovery Vehicles valued at $3M+
- Mentored 4 crews on recovery operations during 3 large-scale training exercises, and completed 50+ recoveries
- Supervised and trained a team of 6 in rigging techniques, expedient repairs, proper towing, lifting, winching, and accounted for $2M worth of equipment
- Supported junior advancement by coordinating 60,000 cumulative hours of military education, leading to 3 promotions

## PROJECTS

**FPGA-Based Interactive System (Nexys A7)** *March 2026 – May 2026*
*Digital Design*

- Designed modular synchronous RTL in Verilog, implementing a six-state finite state machine, sequence memory, timers, and Linear-Feedback Shift Register (LFSR)-based pseudo-random logic
- Developed reliable asynchronous input interfaces using two-stage synchronizers, switch debouncing, and single-cycle pulse generation
- Implemented a 640×480 VGA timing and rendering pipeline from a 100 MHz system clock, defining clock, pin, and I/O constraints in Xilinx Vivado using XDC
- Created self-checking Verilog testbenches for functional verification; synthesized and implemented the integrated design in Xilinx Vivado and performed FPGA hardware bring-up and validation

**Dual-Sensor Speed Trap System** *October 2025 – December 2025*
*Embedded Systems*

- Developed embedded C firmware on an AVR microcontroller using timers, interrupts, ADC, GPIO, EEPROM, and PWM to perform real-time speed measurements with optical sensors
- Implemented low-level device drivers for LCD, rotary encoder, servo motor, RGB LED, and buzzer using direct register access and fixed-point arithmetic under real-time constraints
- Built, tested, and debugged a multi-file embedded project with Makefiles; used an oscilloscope and digital multimeter to validate signal timing, voltage levels, and I/O behavior on physical hardware

## SKILLS

- Languages: C, C++, Python, Go, x86 Assembly, MIPS
- Design & Simulation Tools: Xilinx Vivado, ModelSim/QuestaSim, GTKWave, MATLAB, Simulink
- Hardware Description Languages: Verilog, VHDL
- Systems & Development Tools: Linux, CLI, Git, GitHub, Docker, Telegraf, InfluxDB, Grafana
- Communication Protocols: MQTT, TCP/IP, HTTP, GPIO, I2C, SPI, UART